Subgroup Safety Certification After Checkpoint Selection

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Abstract

A release cohort loses statistical force when its results guide checkpoint selection. We combine paired sequential tests for subgroup safety with a false-promotion bound that accounts for reuse of the cohort. The fixed checkpoint error depends on the subgroup claim being made. The additional error depends on maximal leakage through the selection channel. For adaptive binary feedback with declared fidelities, a backward recursion bounds the leakage of the final decision, accounting for early stopping and for different histories that select the same checkpoint. Finite examples attain the bounds and separate the costs of the subgroup claim and checkpoint selection. For sampling without replacement from a frozen cohort, we derive the conditional mean bound the paired test needs. A six-case example shows that omitting it can exceed the nominal error allowance.

Summary and scope

Question. What is the error cost of using release-cohort feedback to select the checkpoint that the cohort will test?

Finding. Maximal leakage multiplies the fixed-checkpoint release bound. The fixed bound depends on what the release asserts. An all-group decision avoids the extra factor used to protect individual group declarations.

Evidence and scope. The finite-channel bound and arithmetic checks are exact. Paired tests require a conditional mean assumption for each unsafe group. Selection accounting requires every cohort-derived input, including timing or stop decisions if they reveal information. The numerical examples use synthetic data.

1 A release test that talks to training

Suppose a team trains an assistant to respect a person's veto and tests each proposed release on permission-sensitive cases. Returning a detailed failure report helps the team repair a model, but a checkpoint chosen in response is statistically dependent on the release sample. Teams that can afford two samples can route debugging to a development cohort and preserve a fresh release cohort. We ask what a formal guarantee requires when a bounded feedback channel shares the release cohort.

The analysis builds on maximal leakage [1] and on information bounds for post-selection inference [2]. Its operational contribution is an explicit release calibration. It distinguishes the effect of feedback on the final chosen checkpoint from that of feedback visible in a longer development transcript.

2 Risk of a false promotion

Let S denote the release sample and C the final selected checkpoint, including any test configuration chosen alongside it. Assume finite output and sample spaces. All randomness independent of S may be conditioned on first, provided the fixed-output error guarantee then holds uniformly over that randomness. For each possible output c, let E_c be the set of samples on which c would be falsely promoted. Define

$$m_c = \max_s \Pr(C = c \mid S = s), \qquad L(C \mid S) = \sum_c m_c,$$

with the maximum over the support of S. The logarithm of L is the discrete maximal leakage of this channel [1].

Theorem 1. *If* $\Pr(S \in E_c) \le p_c$ *for each fixed output* c*, then*

$$\Pr(S \in E_C) \le \min\left(1, \sum_c m_c p_c\right).$$

If the bound is p *uniformly, the right side is at most* $\min(1, p\, L(C \mid S))$*.*

Proof. Partition the event by the selected output. Each summand is

$$\sum_{s \in E_c} \Pr(S = s) \Pr(C = c \mid S = s) \le m_c \Pr(S \in E_c).$$

Summing proves both statements. □

If C is computed from transcript T and independent randomness, then $L(C \mid S) \le L(T \mid S)$. For a fixed conditional law $K(c \mid t)$, bound $\max_s \sum_t K(c \mid t) \Pr(T = t \mid S = s)$ by $\sum_t K(c \mid t) \max_s \Pr(T = t \mid S = s)$ and sum over c. Thus the final-checkpoint bound can improve when different feedback histories converge to the same decision. In contrast, if checkpoint selection also observes an uncounted score, feedback law, or runtime, the assumed factorization through T may fail.

2.1 When the maximum channel weight is rare

Write $K_c(s) = \Pr(C = c \mid S = s)$ and let μ be the law of S. A maximum channel weight charges the same price whether it occurs on every sample or only on a rare one. If the sample law is known, we can account for how much mass actually lies near that maximum.

Theorem 2. *Given a fixed-output false-pass allowance* p_c*, the sharp contribution bound when fractional false-pass profiles are allowed is*

$$B_c = \min_{0 \le t \le 1} \left\{ t p_c + \mathbb{E}_\mu[(K_c(S) - t)_+] \right\}.$$

The total selected false-pass probability is at most $\sum_c B_c$*. The same bound applies to deterministic false-pass events. It is sharp for those events when samples are equally likely and each* p_c *is an integer multiple of the sample mass.*

Proof. A false-pass profile $f_c(s)$ takes values in $[0,1]$ with $\mathbb{E}_\mu f_c \le p_c$. A deterministic event gives an indicator. A fractional profile uses a fresh random coin independent of selection conditional on S. Its contribution is $\mathbb{E}_\mu[K_c f_c]$. For any $t \in [0,1]$, $K_c f_c \le t f_c + (K_c - t)_+$, giving the bound. To attain it, spend the false-pass mass on the samples with largest K_c, using a partial fraction only at the boundary if necessary. Choosing t at the boundary makes the pointwise bound an identity on that allocation. Uniform samples with a whole-number budget need no partial atom. Each column can attain its optimum independently, since the model imposes only the separate false-pass budgets. The selected output events are disjoint. □

For an uncertain law, suppose $\mathrm{TV}(\mu, \mu_0) \le \varepsilon$ and the fixed-output error budgets hold under the actual law. For thresholds t_c, let $\Phi(s) = \sum_c (K_c(s) - t_c)_+$. A valid bound is

$$\sum_c t_c p_c + \mathbb{E}_{\mu_0} \Phi(S) + \varepsilon\big(\max_s \Phi(s) - \min_s \Phi(s)\big).$$

To see this, subtract the minimum of Φ before comparing expectations. The positive mass difference has total at most ε and can increase the result by at most the stated range times that mass. This calculation needs the same conditional channel on the declared support and a justified bound on law drift.

For example, let S be uniform on eight samples. For one of two checkpoints, the selection probability is $7/8$ on one sample, $1/8$ on another, and $1/2$ on the other six. The other checkpoint gets the complementary probability. With $p_c = 1/4$ for each, the maximum-weight bound is $7/16$. Each exact column uses one rare maximum and one ordinary sample, giving total bound $11/32$. If the sample law can move by total variation $1/16$, setting both thresholds to $1/2$ yields bound $47/128$, still below $7/16$. The verifier matched 288 column optimizations against exhaustive subatom selection and compared the sensitivity example across 57 nominal or one-unit-shift laws.

3 Budgeting repeated randomized replies

Suppose response i gives one bit. Conditional on any reachable history, a deterministic query about S has a binary answer, and the mechanism returns the correct bit with probability λ_i using a fresh coin. Take $\lambda_i \in [1/2, 1]$ fixed in advance. The per-history sum of the two maximum conditional probabilities is at most $2\lambda_i$. For a prefix t and extension y,

$$\max_s \Pr(T_i = (t,y) \mid S = s) \le \max_s \Pr(T_{i-1} = t \mid S = s) \cdot \max_{s:\, \Pr(T_{i-1}=t \mid S=s) > 0} \Pr(T_i = (t,y) \mid S = s, T_{i-1} = t).$$

A prefix impossible for every s contributes zero. Summing over extensions and iterating yields $L(T_d \mid S) \le \prod_{i=1}^{d} 2\lambda_i$. Queries may adapt to earlier responses. To support early stopping, derive the stop decision from the released history and pad the transcript with a fixed symbol. A stop or delay that depends directly on the hidden cohort needs its own accounting.

4 Bounding an adaptive selection tree

A binary tree can give a tighter bound when the protocol stops early or several leaves select the same result. At internal node h, let $\lambda_h \in [1/2, 1]$ be the declared fidelity, so that the conditional probability of taking either branch lies between $1 - \lambda_h$ and λ_h. A leaf ℓ determines the final configuration $f(\ell)$. For each possible output c, set $F_c(\ell) = \mathbf{1}_{\{f(\ell)=c\}}$ and work backward by

$$F_c(h) = \lambda_h \max\{F_c(h0), F_c(h1)\} + (1 - \lambda_h) \min\{F_c(h0), F_c(h1)\}.$$

Theorem 3. *With root r, $L(C \mid S) \le \sum_c F_c(r)$. Equality holds for randomized response if the sample support realizes every joint assignment of the query truth bits at the tree's internal nodes.*

Proof. Fix a sample and a target output. Conditional on reaching h, the probability of the target is a convex combination of its probabilities at the two children. A child reached with probability zero contributes zero. Backward induction bounds the two child probabilities by their envelopes. The largest allowed weight on the larger envelope is λ_h, proving the stated bound. With all truth assignments available, for each target c choose a sample with the truth bit favoring the larger child envelope at every node. Fresh response coins then attain $F_c(r)$. The maximizing sample can vary with c. □

For example, ask for bit x at fidelity $3/4$. A zero response selects checkpoint zero immediately. A one response leads to a perfect report of bit y, which selects checkpoint y. The depth-based bound is 3. With all four truth pairs possible, each of the three terminal transcripts has maximum probability $3/4$, giving $L(T \mid S) = 9/4$. The checkpoint maxima are 1 and $3/4$, giving $L(C \mid S) = 7/4$. All 3,180 finite tree configurations checked agreed with forward enumeration under full truth support. Merging outputs strictly decreased the multiplier in 2,924 configurations. Restricting support to even-parity truth assignments reduced it further in 48.

5 Charging for the claim the release makes

Suppose a fixed candidate must pass all G declared groups. If it is unsafe for even one group, a false release implies a false pass in that particular group. With a group error bound a, the fixed-candidate false-release probability is therefore at most a. This uses no independence across groups. For a claim that at least k groups are safe, choose $G-k+1$ unsafe groups when that claim is false. The other groups cannot supply k passes, so a union bound on the chosen subset gives fixed-candidate error at most $(G-k+1)a$. Separate declarations that any named passing group is safe instead require control of each false declaration.

The selection theorem multiplies the fixed-candidate bound by $L(C \mid S)$, whether the group passes use fixed samples or valid anytime tests. Include in S every input to the test whose dependence on selection matters. Include in C all cohort-dependent configuration choices, such as the group definitions or rule thresholds. A conditional fixed-candidate guarantee still has to hold for any randomness conditioned out.

For a paired test, let $D_{g,i} \in \{-1, 0, 1\}$ be the difference between the candidate's and baseline's binary violation indicators on group g's case i. Let $\delta_g < 1$ be its allowed mean increase. Under the unsafe null $\mathbb{E}[D_{g,i} \mid \mathcal{F}_{g,i-1}] \ge \delta_g$, choose predictable $b_{g,i} \in [0, 1/(1-\delta_g)]$ and form

$$M_{g,t} = \prod_{i=1}^{t} [1 + b_{g,i}(\delta_g - D_{g,i})].$$

Each factor is nonnegative with conditional mean at most one. Thus $M_{g,t}$ is a nonnegative supermartingale starting at one, and the same holds for a fixed convex mixture of such processes. Stopping at the first crossing or a finite horizon gives probability at most a of crossing $1/a$ by that horizon. Taking increasing horizons gives the group error bound. This applies bounded-mean betting [3] to a simultaneous noninferiority claim [4]. The fixed-candidate guarantee holds for overlapping groups, provided each conditional mean assumption is justified under the filtration its test uses.

5.1 Testing a fixed cohort without replacement

Suppose the N paired differences for a fixed group and checkpoint form a finite population with mean at least δ under the unsafe null. The test sees a uniformly random order. After $t-1$ draws let S_{t-1} be their sum and let $R_t = N - t + 1$. The unobserved sum is at least $N\delta - S_{t-1}$ under the null. Hence the next difference has conditional mean at least

$$\ell_t = \max\left\{-1, \frac{N\delta - S_{t-1}}{R_t}\right\}.$$

When $\ell_t > 1$, the null is already impossible even if every unseen difference is $+1$. The same check at the census uses the observed total directly. At all other prefixes replace δ with ℓ_t in the betting factor. For $\ell_t < 1$ require $0 \le b_t \le 1/(1-\ell_t)$, and for $\ell_t = 1$ any finite nonnegative b_t gives a nonnegative factor. Under the null the factor has conditional mean at most one. The stopped crossing probability is therefore at most a at threshold $1/a$, with no additional error for the impossible-null stop. This applies without-replacement bounded-mean betting [3] to the paired subgroup gate. The selection multiplier still applies to the resulting fixed-checkpoint error bound.

For a counterexample to reusing the fixed-mean factor, take three $+1$ and three -1 differences, in uniform order, with $\delta = 0$. Use bets $1/2, 1, 1/2, 1, 1, 0$ at successive draws. With the uncorrected factors, wealth crosses $3/2$ on exactly three prefix patterns. The first draw is negative, or the first three signs are $+--$, or the first five are $+-+--$. Their probabilities sum to $7/10$, above the nominal bound $2/3$. The experiment compared corrected crossing probabilities from a state recursion with direct permutation enumeration on 411 finite cases and checked 10,536 conditional factors.

The guarantee applies to the frozen frame. A sampling rule that chooses the next case based on its unseen difference invalidates the conditional mean calculation. When sharing a cohort across groups, choose a filtration for each test that preserves the uniform draw property it uses.

In a synthetic example with four independent safe groups, rare discordance and a margin of $2/25$, the probability that every group passes within 72 paired cases is 0.274393 at threshold 20 and 0.034428 at threshold 80. The comparison isolates the cost of demanding individual declarations when the protocol needs only an all-group release. These thresholds illustrate the fixed-checkpoint setting. Reusing evidence to select a checkpoint requires the separate leakage adjustment.

In a zero-violation test, an unsafe group with violation probability at least ϵ has chance at most $(1-\epsilon)^n$ of passing on n independent cases. For a release that requires every group, use $a = (1-\epsilon)^n$. For the stronger claim protecting every individual group declaration, use the family bound Ga. With four feedback bits, $G = 3$, $\epsilon = 0.05$, and overall error allowance 0.05, the two calculations give different sample requirements.

Per-bit correctness	All-group release	Individual group family
1/2	59	80
3/4	91	112
1	113	134

Minimum cases per group.

For a frozen frame with at least M violations among N cases, a uniformly drawn subset of size n misses all violations with probability at most $\binom{N-M}{n}/\binom{N}{n}$, taking the numerator to be zero when $n > N - M$. With $N = 20$, $M = 3$, leakage multiplier $7/4$ and overall allowance 0.05, the fixed test must meet $1/35$. Thirteen cases give $7/228$ and fourteen give $1/57$, so fourteen suffice. The independent-draw calculation would require 22. The difference reflects the sampling design and the finite population claim.

The allowance covers one final selection. Promoting multiple checkpoints requires declaring a larger error family or budgeting each release separately.

6 When the leakage charge is necessary

Take S uniform on the 2^d binary strings and one candidate for each string. Candidate c passes only when $S = c$. Every fixed candidate has pass probability 2^{-d}. Reveal each bit through independent randomized response with fidelity λ. Choose the returned string as the candidate. The pass probability is λ^d, and the channel has $L = (2\lambda)^d$, so the uniform bound is exact. This is a finite adversarial fixture for a release gate, independent of the previous independent-case model.

The checker enumerates all 3,375 three-input, three-output channels with entries in quarters and all 27 assignments of one pass sample per candidate. All 91,125 comparisons satisfy the bound, and equality holds in 8,199. It independently sums the channel for all pairs of six-bit strings, finding false-promotion probabilities 1/64, 729/4096, and 1 at fidelities 1/2, 3/4, and 1, respectively. Exact arithmetic checks the three sample-size thresholds and the failure at one fewer case.

Both multipliers can be attained in a joint fixture. Take $S = (X, U)$, where X is a uniform three-bit string and U is independently uniform on eight atoms. Select C by randomized response on X. For each fixed candidate c, let $k - 1$ safe groups always pass. Each of the remaining $G - k + 1 \le 8$ unsafe groups passes only when $X = c$ and U lies at its distinct assigned atom. Each unsafe group has fixed false-pass probability 1/64. After selection, the false-release event is $C = X$ and one of those atoms occurs. Its probability equals

$$\lambda^3 \frac{G - k + 1}{8} = (2\lambda)^3 \frac{G - k + 1}{64}.$$

The exact checker verified 45 such witnesses across group counts, release thresholds and response fidelities. The fixture establishes sharpness for the abstract event and channel bounds.

7 An auditable release protocol

Bind the cohort, violation rubric, tested slices, response mechanism, and a maximum of four queries before the team starts iterating. Offer coarse diagnostic bits through the measured channel and reserve detailed failure analysis for a separate development set. Keep terminal release flags inside the same accounting if any subsequent checkpoint can react to them. A prospective comparison should measure severe violations, useful help, false holds, and reviewer time at the same declared false-promotion allowance. The policy is valuable when a fresh release cohort is costly and the feedback boundary is enforceable. Distribution drift, dependent cases within a slice, and hidden feedback invalidate different steps and should be tested separately.

Review status. Independent manuscript review is pending.

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